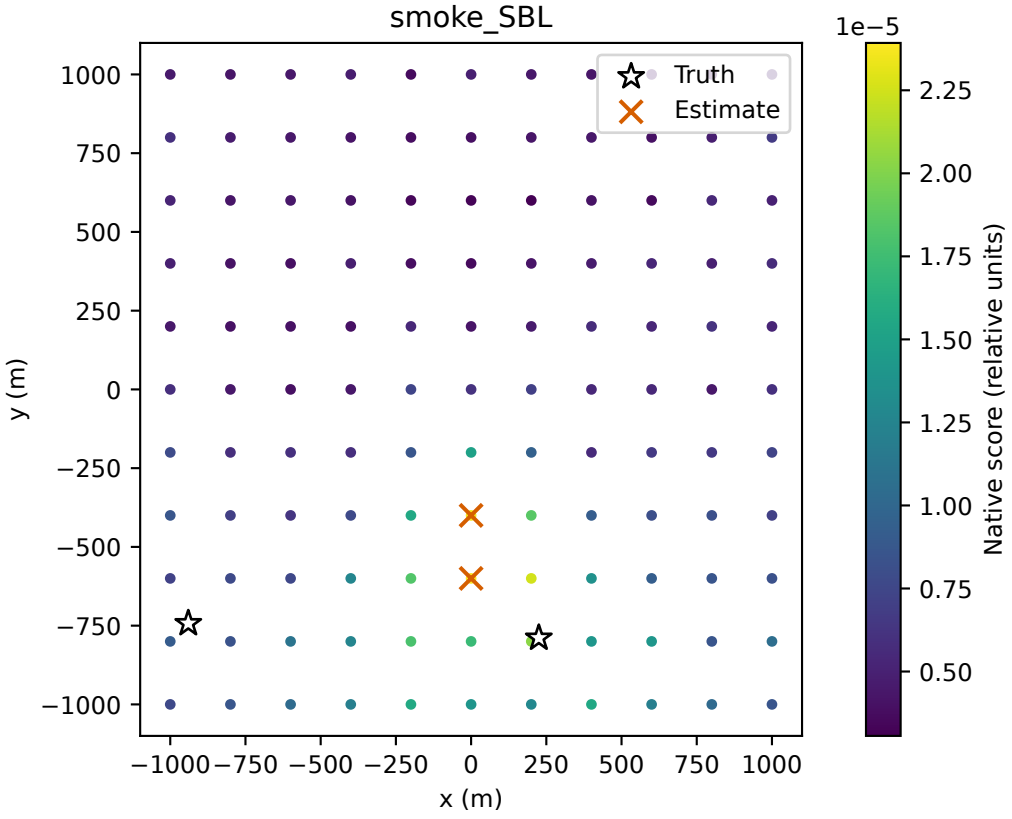ENGINEERING SMOKE / NOT PERFORMANCE